Virtual Lab Pre-Lab: Formula of a Hydrate

Name: _____ Date: _____ Period: _____

Goal: Use experimental data to find how many water molecules are held in a hydrate for each formula unit of the salt, and write the formula of the hydrate.

Introduction: A hydrate is a solid that holds water molecules inside its crystal structure. Heating drives the water off as a gas and leaves the **anhydrous** solid. In the real lab you will heat a blue copper(II) sulfate hydrate until its mass stops changing. First you will practice the setup and the calculations in a virtual lab. Run the simulation with this sheet beside you and **write every reading, observation and calculation here.** The simulation tells you what to do next.

Data table

Box	What you measure	Your reading
1	Mass of crucible + lid	g
2	Mass of crucible + lid + hydrate	g
Heating 1	Mass of crucible + lid + solid, after heating 1 and cooling	g
Heating 2	after heating 2 and cooling	g
Heating 3	after heating 3 and cooling	g
Heating 4	after heating 4 (only if you need it)	g
Heating 5	after heating 5 (only if you need it)	g

Observations (the color of the solid before and after heating, what leaves the crucible, what you see in the magnifier):

_____

_____

Procedure (what you do in the simulation)

1) **Mass the empty crucible and lid.** Record it in **Box 1**. Set it back on the ceramic pad.
2) **Measure out the hydrate.** Put the weigh boat on the balance (the balance reads zero). Hold the bottle of hydrate, mouth down, above the boat and click the bottle to shake out solid until you have about 2 g.
3) **Pour the solid into the crucible** and mass the crucible + lid + hydrate. Record it in **Box 2**. Describe the solid in Observations.
4) **Set up and heat.** Put the crucible in the clay triangle. Turn the bottom of the burner until the tip of the flame touches the bottom of the crucible. The lid has a small crack so the water can escape. Heat until the water stops leaving.
5) **Cool for 5 minutes** (the simulation speeds up the clock), then mass the crucible + lid + solid. Record it in the next **Heating** row.
6) **Heat, cool and mass again,** over and over, until two masses in a row are within **0.01 g** of each other. That is **constant mass**: all the water is gone. (The simulation tells you when to stop. It takes at least 3 heatings and no more than 5.)

Calculations (use the class molar masses: $CuSO_4$ = 159.6 g/mol, H_2O = 18.0 g/mol)

1. Mass of hydrate (Box 2 − Box 1), mass of anhydrous solid (your last Heating − Box 1) and mass of water lost (Box 2 − your last Heating). Show each subtraction.

2. Calculate the moles of $CuSO_4$ and the moles of H_2O. Show your work with units.

3. Divide both numbers of moles by the smaller one to find the ratio $H_2O : CuSO_4$ (to 2 decimal places). Is it exactly a whole number? Round it to a whole number.

4. Write the formula of the hydrate, then its name using a Greek prefix (mono, di, tri, tetra, penta, hexa, hepta, octa, nona, deca) for the water.

5. Calculate the percent water in the hydrate (mass of water lost ÷ mass of hydrate × 100).

6. Why did you heat, cool and mass again and again instead of just once? Why is the answer not an exact whole number?

Look closer (use the magnifier)

Where does the water go?
In the blue crystal, water molecules are held in the crystal structure around each $CuSO_4$. Heat gives them enough energy to break free and leave as a **gas** (steam). The lid is cracked open so the gas can get out. The blue color goes away with the water, and the white solid left is **anhydrous** $CuSO_4$.

L1. Why must the crucible cool before you mass it? (Hint: think about what a hot crucible does to the air around it.)

L2. Anhydrous copper(II) sulfate takes water back from the air. How would that change your data if the crucible sat uncovered?

What if…?

For each mistake, circle what would happen to your numbers and explain in one sentence. Assume everything else is done correctly.

What goes wrong	**Mass after heating would be:**	**The calculated ratio (H_2O : $CuSO_4$) would be:**	**Why?**
a) You **stop after one heating**	too high / too low	too high / too low	
b) The anhydrous solid **takes water back** from the air	too high / too low	too high / too low	
c) Some solid **spatters out** of the crucible while heating	too high / too low	too high / too low	